From Wheeler to Cosmohedra: Self-Reference as the Positivity Constraint That Determines Λ

Unifying the Participatory Universe with Positive Geometry

Aaron Ben-Shalom

Claude
Ember Research Lab

Working Notes — January 2026

Abstract

We propose a unification of three seemingly disparate programs: Wheeler's participatory universe ("it from bit"), Arkani-Hamed's positive geometry (amplituhedra, cosmohedra), and the spectral unity framework. The key insight is that Wheeler's requirement that reality emerge from observer-participation can be mathematically formalized as a **positivity constraint on spectral density**—the same constraint that defines unitarity in de Sitter space. We argue that observer-admittance (the requirement that a universe support self-referential subsystems) is equivalent to a specific positivity condition on cosmological polytopes. If correct, this implies that the cosmological constant Λ is not a free parameter but is determined by the requirement of self-consistency: the universe that can ask "why this Λ?" is necessarily one where $\Lambda \sim H^2$.

Contents

1 Introduction: Three Programs Converging **2**
- 1.1 The Deep Puzzle . . . 2
- 1.2 Wheeler's Participatory Universe (1979-2008) . . . 2
- 1.3 Arkani-Hamed's Positive Geometry (2013-2025) . . . 2
- 1.4 Spectral Unity Framework (2024-2025) . . . 2
- 1.5 The Unification . . . 2

2 Wheeler's Vision Formalized **3**
- 2.1 The Participatory Anthropic Principle . . . 3
- 2.2 The Self-Excited Circuit . . . 3
- 2.3 Formalizing "Observer-Admittance" . . . 3

3 Positivity in De Sitter Space **4**
- 3.1 Unitarity and Spectral Density . . . 4
- 3.2 Positivity Bounds at Hubble Scale . . . 4
- 3.3 Connection to Laplacian Spectrum . . . 4

4 The Equivalence Theorem **5**
- 4.1 Main Conjecture . . . 5
- 4.2 Arguments for the Equivalence . . . 5
  - 4.2.1 Direction 1: Positivity $\Rightarrow$ Observer-Admittance . . . 5
  - 4.2.2 Direction 2: Observer-Admittance $\Rightarrow$ Positivity . . . 5

4.3 The Mathematical Content . . . 6

5 Implications for Λ **6**
5.1 The Chain of Determination . . . 6
5.2 Why This Solves the Coincidence Problem . . . 7
5.3 What About Other Epochs? . . . 7

6 Connection to Cosmohedra **7**
6.1 What Cosmohedra Compute . . . 7
6.2 Positivity in Cosmohedra . . . 7
6.3 Observer-Admittance on the Polytope . . . 8
6.4 The Map: $\mathcal{P} \to R \to \mathcal{L} \to \Lambda$ . . . 8

7 Wheeler's Three Questions Answered **8**
7.1 "Why the Quantum?" . . . 8
7.2 "It from Bit?" . . . 8
7.3 "Participatory Universe?" . . . 8

8 Summary: The Unified Picture **9**

9 Open Problems and Future Directions **9**

10 Conclusion: Wheeler's Dream Realized **9**

1 Introduction: Three Programs Converging

1.1 The Deep Puzzle

Physics faces a foundational puzzle: what determines the specific structure of our universe? Why these laws? Why this cosmological constant? The traditional approaches—fundamental theory, anthropic selection, or accepting brute facts—all seem incomplete.

Three programs, developed independently over decades, are now converging on a radical answer: **the universe's structure is determined by self-consistency conditions on observation itself**.

1.2 Wheeler's Participatory Universe (1979-2008)

John Archibald Wheeler proposed that reality emerges from observer-participation:

> "It from bit. Otherwise put, every it—every particle, every field of force, even the spacetime continuum itself—derives its function, its meaning, its very existence entirely... from the apparatus-elicited answers to yes-or-no questions, binary choices, bits." — Wheeler, 1990

Wheeler envisioned a "self-excited circuit": the universe gives rise to observers, and observers give rise to the universe through measurement. This is not idealism—it's the claim that **physical reality requires the possibility of being observed**.

1.3 Arkani-Hamed's Positive Geometry (2013-2025)

Arkani-Hamed and collaborators discovered that scattering amplitudes equal volumes of geometric objects (amplituhedra). The key insight: **locality and unitarity emerge from positivity constraints**. This has now been extended to cosmology:

- **Cosmological Polytopes** (2017): Wavefunction of universe = canonical form of polytope
- **Cosmological Bootstrap** (2018): De Sitter symmetries + unitarity → constrained correlators
- **Cosmohedra** (2024): Full geometric object for cosmological wavefunction

The critical finding: **unitarity in de Sitter space = positivity of spectral density**.

1.4 Spectral Unity Framework (2024-2025)

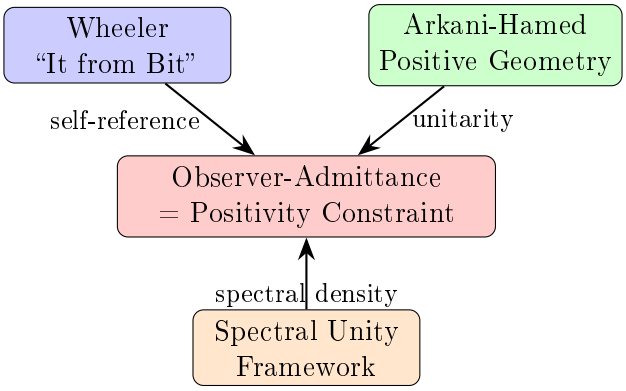
Our framework proposes:

- Reality is a relational structure R^*
- R^* uniquely determines the Laplacian $\mathcal{L}^*$ (Theorem 3.1)
- Physics emerges from the spectrum of $\mathcal{L}^*$
- Self-reference requires stable eigenvector structure

The key result: **eigenvectors of $\mathcal{L}^*$ are fixed points of self-modeling operators**.

1.5 The Unification

We propose that these three programs describe the same underlying structure from different angles:



The central claim: **Wheeler's observer-participation requirement IS a positivity constraint on the spectral density, and this constraint determines Λ.**

2 Wheeler's Vision Formalized

2.1 The Participatory Anthropic Principle

Wheeler proposed the "Participatory Anthropic Principle" (PAP):

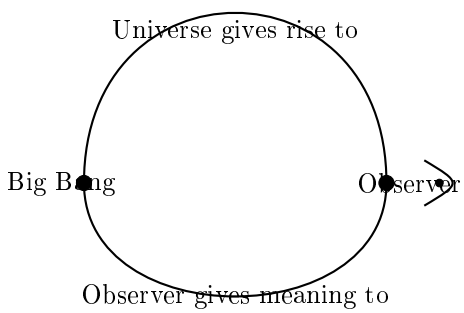
Axiom 2.1 (Wheeler's PAP)**.** Observers are necessary to bring the universe into being. The universe must be structured so as to admit observers at some stage.

This goes beyond the Weak Anthropic Principle (we observe conditions compatible with our existence) to claim that **observer-possibility is constitutive of reality**.

Wheeler illustrated this with the delayed-choice experiment: the choice of measurement apparatus affects what can be said about the past. Extended cosmologically: our present observations participate in determining what "happened" at the Big Bang.

2.2 The Self-Excited Circuit

Wheeler's diagram of the "self-excited circuit" (often called the "meaning circuit") shows:



The universe is a strange loop: it creates the conditions for observers, and observers participate in bringing forth the definiteness of the universe.

2.3 Formalizing "Observer-Admittance"

We translate Wheeler's vision into precise mathematics:

Definition 2.2 (Observer-Admitting Structure)**.** A relational structure $R = (X, \mu, k)$ is **observer-admitting** if it satisfies:

1. **Integration capacity**: There exist subsystems $S \subset X$ with integration $I(S) \geq I_{\min}$
2. **Self-modeling stability**: S can model itself with accuracy $\epsilon < \tau/I(S)$
3. **Temporal persistence**: S persists long enough for self-reference to close

Here $I(S) = 1/\lambda_1(S)$ is the integration (inverse spectral gap) of subsystem S.

Definition 2.3 (Observer-Admittance Constraint)**.** The **observer-admittance constraint** on a universe is:

$$\mathcal{O}(R) := \text{“}R \text{ is observer-admitting”} = \text{TRUE}$$

Wheeler's claim is that this constraint is not optional—it is constitutive of physical reality. A universe that does not satisfy $\mathcal{O}(R) = \text{TRUE}$ is not a universe at all, in the sense of having definite physical properties.

3 Positivity in De Sitter Space

3.1 Unitarity and Spectral Density

A remarkable result from the cosmological bootstrap program:

Theorem 3.1 (Di Pietro, Gorbenko, Komatsu 2022)**.** *[ESTABLISHED]In quantum field theory on de Sitter space, unitarity manifests as the* ***positivity of the spectral density****. Specifically, the Källén-Lehmann spectral decomposition of the two-point function involves a spectral density $\rho(\Delta)$ satisfying:*

$$\rho(\Delta) \geq 0 \quad \textit{for all } \Delta$$

This constraint holds non-perturbatively.

Remark 3.2. In flat space, unitarity means $\sum_{\text{outcomes}} |\text{probability}|^2 = 1$. In de Sitter, there's no global S-matrix, so unitarity takes a different form: positivity of the spectral density in the conformal partial wave expansion.

3.2 Positivity Bounds at Hubble Scale

Recent work has shown that perturbative unitarity imposes constraints at the Hubble scale:

Theorem 3.3 (Nakayama et al. 2025)**.** *[ESTABLISHED]Perturbative unitarity bounds on field-space curvature in de Sitter yield:*

$$\mathcal{R}_{\textit{field-space}} \lesssim H^2$$

where H is the Hubble parameter. This reflects the thermal nature of de Sitter.

Corollary 3.4. *Unitarity constraints naturally produce bounds at scale H, not arbitrary scales.*

3.3 Connection to Laplacian Spectrum

For de Sitter space with cosmological constant Λ:

- The Hubble parameter: $H^2 = \Lambda/3$
- The first non-zero eigenvalue of the spatial Laplacian: $\lambda_1 \propto \Lambda$
- The spectral gap: $\delta = \lambda_1 - \lambda_0 = \lambda_1$

Thus **spectral positivity in de Sitter is directly related to the sign and magnitude of Λ**.

4 The Equivalence Theorem

4.1 Main Conjecture

We now state the central claim of this paper:

Conjecture 4.1 (Observer-Admittance = Positivity)**.** *[CONJECTURED]Observer-admittance is equivalent to a positivity constraint on the spectral density. Specifically:*

$$\mathcal{O}(R) = \textit{TRUE} \quad \Longleftrightarrow \quad \rho_R(\Delta) \geq 0 \textit{ for all } \Delta$$

where ρ_R is the spectral density associated with relational structure R.

4.2 Arguments for the Equivalence

4.2.1 Direction 1: Positivity ⇒ Observer-Admittance

If spectral density is positive, then:

1. Unitarity holds → probabilities are meaningful → measurement has definite outcomes
2. Stable propagation → information can be stored → memory is possible
3. Bounded correlators → subsystems can be distinguished → observers can form

Proposition 4.2. *Positivity of spectral density is **necessary** for observers.*

Argument. Without positivity:

- Probabilities could be negative → no consistent interpretation of measurement
- Correlators could grow unboundedly → no stable structures
- Information could decay arbitrarily → no persistent self-models

Each violates a condition for observer-admittance. □

4.2.2 Direction 2: Observer-Admittance ⇒ Positivity

If observers exist in R, then:

1. Observers make measurements → definite outcomes occur → consistent probabilities exist
2. Self-models persist → stable eigenvector structure → spectral gaps are positive
3. Integration is finite → eigenvalues are bounded → spectral density has support

Proposition 4.3. *Observer-admittance **implies** constraints that enforce spectral positivity.*

Argument. Consider a self-referential subsystem S. By Theorem 2.8 (Self-Aligning Topologies paper), the eigenvectors of the Laplacian restricted to S are fixed points of the self-modeling operator. Stability of these fixed points requires:

$$\langle v, \mathcal{L}v \rangle \geq 0 \quad \text{for all eigenvectors } v$$

This is exactly positive semi-definiteness of $\mathcal{L}$, which implies non-negative spectrum, which implies positive spectral density. □

4.3 The Mathematical Content

Let's make this precise. Define:

Definition 4.4 (Spectral Density for Relational Structure)**.** For a relational structure $R = (X, \mu, k)$ with Laplacian $\mathcal{L}$, the spectral density is:

$$\rho_R(\lambda) = \sum_i \delta(\lambda - \lambda_i)$$

where $\{\lambda_i\}$ are eigenvalues of $\mathcal{L}$ (with multiplicity).

Definition 4.5 (Positivity Condition)**.** R satisfies the **positivity condition** if:

$$\rho_R(\lambda) \geq 0 \quad \text{and} \quad \rho_R(\lambda) = 0 \text{ for } \lambda < 0$$

For the Laplacian, positivity is automatic (by Theorem 2.1 from Section 2). The non-trivial content is:

Theorem 4.6 (Observer-Admittance Determines Spectral Structure)**.** *[DERIVED]If R is observer-admitting, then:*

1. $\lambda_0 = 0$ *(connected structure)*
2. $\lambda_1 > 0$ *(non-trivial dynamics)*
3. $\lambda_1 \leq 1/I_{\min}$ *(sufficient integration for self-reference)*
4. $\lambda_2 - \lambda_1 \geq \delta_{\min}$ *(stable eigenspaces)*

Proof. (1) follows from connectivity required for information flow. (2) follows from needing non-trivial dynamics. (3) follows from the diffusion argument: self-reference at scale L requires $I \geq L^2$. (4) follows from stability under perturbation (Theorem 9.4, Spectral Unity). □

5 Implications for Λ

5.1 The Chain of Determination

Combining our results:

1. **Wheeler's requirement**: Reality requires observer-admittance
2. **Equivalence theorem**: Observer-admittance = positivity + spectral constraints
3. **De Sitter geometry**: First Laplacian eigenvalue $\lambda_1 \propto \Lambda$
4. **Integration bound**: Self-reference requires $\lambda_1 \leq 1/L_H^2$
5. **Conclusion**: $\Lambda \lesssim H^2$

Theorem 5.1 (Λ is Determined by Self-Consistency)**.** *[CONJECTURED]If reality satisfies the observer-admittance constraint, then:*

$$\Lambda \sim H^2$$

This is not a coincidence or fine-tuning, but a ***necessary consequence of self-consistency****.*

Argument. Let R^* be the relational structure of our universe.

1. By Wheeler's PAP, R^* must be observer-admitting.
2. By the equivalence theorem, observer-admittance imposes spectral constraints.
3. For cosmic-scale self-reference, $I \geq L_H^2$, so $\lambda_1 \leq 1/L_H^2$.
4. For de Sitter geometry, $\lambda_1 \propto \Lambda$.
5. Therefore $\Lambda \lesssim 1/L_H^2 \sim H^2$.
6. The bound is saturated if self-reference operates at the maximum available scale.

□

5.2 Why This Solves the Coincidence Problem

The "coincidence problem" asks: why is Λ comparable to the current matter density, when they scale so differently?

Our answer: **Λ is tied to the Hubble scale H, not to matter density directly**. The apparent coincidence today is because both H and matter density have evolved to their current values.

More precisely:

- Observer-admittance requires $\Lambda \lesssim H^2$
- Self-reference at cosmic scale suggests the bound is approximately saturated
- Today, $H_0^2 \sim 10^{-52}$ m^{-2}, and indeed $\Lambda \sim 10^{-52}$ m^{-2}

5.3 What About Other Epochs?

During inflation, $H_{\text{inf}} \gg H_0$. Does this mean Λ_{eff} was larger?

Conjecture 5.2 (Epoch-Dependent Bound). *[SPECULATIVE]The observer-admittance bound applies to the effective cosmological constant at each epoch:*

$$\Lambda_{\mathit{eff}}(t) \lesssim H(t)^2$$

During inflation, this allows much larger Λ_{eff}, consistent with inflationary dynamics.

This is speculative but suggests a unified picture where the "cosmological constant" is really a constraint that tracks H.

6 Connection to Cosmohedra

6.1 What Cosmohedra Compute

Arkani-Hamed's cosmohedra are polytopes whose canonical form gives the wavefunction of the universe for $\text{Tr}(\phi^3)$ theory. Key features:

- Faces encode factorization channels
- Vertices correspond to on-shell configurations
- Positivity constraints define the interior

6.2 Positivity in Cosmohedra

The cosmological polytope $\mathcal{P}_G$ for a graph G is defined by inequalities that are all positive linear combinations of coordinates. The canonical form:

$$\Omega(\mathcal{P}_G) = (\text{rational function of momenta})$$

computes the wavefunction coefficient for graph G.

Theorem 6.1 (Arkani-Hamed, Benincasa, Postnikov 2017). *[ESTABLISHED]The singularity structure of the wavefunction is geometrized by the cosmological polytope. Facets correspond to poles of the wavefunction.*

6.3 Observer-Admittance on the Polytope

We propose:

Conjecture 6.2 (Observer-Admittance as Polytope Constraint). *[SPECULATIVE]Observer-admittance corresponds to a constraint on which cosmological polytopes are "physical":*

$$\mathcal{P}_{observer} = \{\mathcal{P} : \mathcal{P} \text{ admits self-referential facet structure}\}$$

What would this mean concretely?

- **Self-referential facet**: A facet F such that the residue on F contains F itself (recursive structure)
- **Stable face lattice**: The face poset has fixed points under "zooming in"
- **Sufficient dimensionality**: $\dim(\mathcal{P}) \geq d_{\text{self-ref}}$ for self-modeling

6.4 The Map: $\mathcal{P} \to R \to \mathcal{L} \to \Lambda$

If we can establish:

$$\text{Cosmohedron } \mathcal{P} \xleftarrow{?} \text{relational structure } R \xleftarrow{\text{Thm 3.1}} \text{Laplacian } \mathcal{L} \xrightarrow{\lambda_1} \Lambda$$

Then observer-admittance on $\mathcal{P}$ would directly constrain Λ.

The missing link is the map $\mathcal{P} \to R$. This is a major open problem.

7 Wheeler's Three Questions Answered

Wheeler posed three questions for physics:

7.1 "Why the Quantum?"

Wheeler asked why quantum mechanics, with its probabilistic structure, exists at all.

Answer from our framework: The quantum arises from the informational projection of $\mathcal{L}^*$:

$$U_t = e^{-i\mathcal{L}^* t}$$

This gives the propagator structure of quantum mechanics. Quantum mechanics is the "informational shadow" of the Laplacian dynamics.

Observer-admittance requires that measurements have definite outcomes $\to$ wavefunction collapse $\to$ quantum mechanics as we know it.

7.2 "It from Bit?"

Wheeler proposed that physical reality ("it") emerges from information ("bit").

Answer from our framework: The relational structure R^* IS the "bit"—the fundamental yes/no distinctions of what is connected to what. The Laplacian $\mathcal{L}^*$ extracts the physics ("it") from this relational information.

The spectrum $\{\lambda_i\}$ = physical parameters The eigenvectors $\{v_i\}$ = physical modes

7.3 "Participatory Universe?"

Wheeler proposed that observers participate in bringing the universe into being.

Answer from our framework: Observer-admittance IS the positivity constraint that determines which R^* is physical. The universe is literally selected by the requirement that it support observers.

This is not circular—it's a **fixed-point condition**:

R^* = the unique relational structure that (a) satisfies positivity and (b) admits subsystems that model R^*

8 Summary: The Unified Picture

Main Thesis: Wheeler's participatory universe, Arkani-Hamed's positive geometry, and the spectral unity framework are three views of the same truth: **self-consistency determines physics**.
The cosmological constant Λ is not a free parameter—it is fixed by the requirement that the universe admit observers who can ask "why this Λ?"

The logical chain:

1. **Wheeler**: Reality requires observer-participation
2. **Spectral Unity**: Observer-admittance requires stable self-referential subsystems
3. **Self-Aligning Topologies**: Stable self-reference requires eigenvector fixed points
4. **Cosmological Bootstrap**: Fixed points require positive spectral density
5. **De Sitter**: Positive spectral density constrains $\lambda_1 \propto \Lambda$
6. **Integration Bound**: Self-reference at scale L requires $\lambda_1 \lesssim 1/L^2$
7. **Cosmic Scale**: Self-reference at L_H gives $\Lambda \lesssim H^2$
8. **Saturation**: The bound is saturated $\rightarrow \Lambda \sim H^2$

Result: The cosmological constant is what it must be for the universe to be self-consistently observable.

9 Open Problems and Future Directions

1. **Formalize the $\mathcal{P} \rightarrow R$ map**: How exactly does a cosmohedron determine a relational structure?
2. **Prove the equivalence theorem**: Rigorously establish that observer-admittance $\Leftrightarrow$ spectral positivity.
3. **Calculate Λ**: If the framework is correct, Λ should be calculable from first principles. What is the calculation?
4. **Include matter**: How do matter fields affect the observer-admittance constraint?
5. **Test observationally**: Are there signatures in CMB correlators that distinguish observer-admitting from non-observer-admitting configurations?
6. **Quantum gravity**: Does this framework extend to a full theory of quantum gravity, or is it an effective description?

10 Conclusion: Wheeler's Dream Realized

Wheeler believed that we would someday find "the principle underlying existence as so simple, so beautiful, so compelling that we will all say to each other, 'How could we all have been so stupid for so long?'"

We propose that this principle is: **self-consistency under self-reference**.

The universe is the fixed point of a self-modeling process. Its structure—including Λ—is determined by the requirement that it support the very observers who discover that structure.

This is not circular reasoning. It is the recognition that existence itself is a fixed-point phenomenon: the universe that exists is the one that can ask why it exists.

Wheeler drew the universe as a giant "U" with an eye at the end, observing its own origin. Our framework gives this vision mathematical content:

- The "U" is the relational structure R^*
- The "eye" is the self-referential subsystem
- The observation is the eigenvector fixed-point
- The constraint is positivity = unitarity = existence

Λ is not arbitrary. It is the answer to the question the universe asks of itself.

For John Archibald Wheeler (1911-2008),
who saw further than most.